

Yohan Jun

Investigator @ MGH, Instructor & Faculty Member in Radiology @ Harvard Medical School

Athinoula A. Martinos Center for Biomedical Imaging, Bldg 149 13th St Rm 2301, Charlestown MA 02129

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Research Areas

Advanced Neuroimaging with MRI	Accelerated MRI, Diffusion MRI, Quantitative MRI
Computational Algorithms for Medical Imaging	Inverse Problem, MR Image Reconstruction, Self-Supervised/Zero-Shot Learning
AI for Automatic Diagnosis of Brain Disorders	Automatic Diagnosis of Brain Tumors Using Deep Learning Algorithms

Education

Ph.D. in Electrical & Electronic Engineering

Seoul, S.Korea

— Yonsei University

Feb. 2022

- Advisor: Dr. Dosik Hwang
- Thesis: "Model-based Deep Learning Reconstruction Methods for Fast Magnetic Resonance Imaging"
- Scholarship: Brain Korea 21 Plus Outstanding Student Fellow Scholarship of Korea Research Foundation
- Award: Best Graduate Student Paper Award

B.S. in Electrical & Electronic Engineering

Seoul, S.Korea

— Yonsei University

Feb. 2016

- Scholarship: National Scholarship for Science & Engineering of Korea Student Aid Foundation

Professional Experience

Instructor in Radiology, Harvard Medical School

Boston, US

- Investigator, Martinos Center for Biomedical imaging, Massachusetts General Hospital
- Instructor and Faculty Member in Radiology, Harvard Medical School
- Pediatric Imaging Research Center, Massachusetts General Hospital

Nov. 2024 - Now

Research Fellow in Radiology, Harvard Medical School

Boston, US

- Postdoctoral Research Fellow, Martinos Center for Biomedical Imaging, Massachusetts General Hospital
- Advisors: Dr. Berkin Bilgic, Dr. Michael S Gee

Mar. 2022 - Nov. 2024

Research Assistant, Yonsei University

Seoul, S.Korea

- Research Assistant, Medical Artificial Intelligence Lab.
- Advisor: Dr. Dosik Hwang

Jan. 2016 - Feb. 2022

Research Intern, Philips Korea

Seoul, S.Korea

- Research Intern with Medical Artificial Intelligence Lab, Yonsei University

Oct. 2017 - Dec. 2017

Teaching Experience

Yonsei University

Seoul, S.Korea

Guest Lecturer, Teaching Assistant

Sep. 2021 - Dec. 2021

- **Introduction Artificial Intelligence**
 - Presented a lecture on principles of deep learning and convolutional neural networks

Guest Lecturer, Teaching Assistant

Mar. 2021 - Jun. 2021

- **Medical Imaging Artificial Intelligence**
 - Presented a lecture on MR image reconstruction using deep learning methods

Guest Lecturer, Teaching Assistant

Sep. 2020 - Dec. 2020

- **Medical Artificial Intelligence**
 - Presented a lecture on principles of MRI and reconstruction methods for fast MRI

Teaching Assistant

Mar. 2018 - Jun. 2018

- **Introduction to Bioengineering for Electrical and Electronic Engineering**

Teaching Assistant

Mar. 2017 - Jun. 2017

- **Electrical and Electronic Engineering Capstone Design**

Honors & Awards

INTERNATIONAL

2024	ISMRM Junior Fellow , The ISMRM 32nd Annual Meeting	Singapore
2024	1st Place Winner, Best Oral Presentation , The ISMRM 32nd Annual Meeting, Diffusion Study Group	Singapore
2024	ISMRM Annual Meeting Program Committee (AMPC) Selected Abstract (Top 1%) , The ISMRM 32nd Annual Meeting	Singapore
2024	ISMRM Summa Cum Laude , The ISMRM 32nd Annual Meeting	Singapore
2022-2023	Distinguished Reviewer , IEEE Transactions on Medical Imaging (IEEE TMI)	
2023	ISMRM Summa Cum Laude , The ISMRM 31st Annual Meeting	Toronto, Canada
2021	1st Rank , Cross-Modality Domain Adaptation for Medical Image Segmentation (crossMoDA-2021 challenge)	Virtual Conference
2021	ISMRM Magna Cum Laude (1) , The ISMRM 29th Annual Meeting	Virtual Conference
2021	ISMRM Magna Cum Laude (2) , The ISMRM 29th Annual Meeting	Virtual Conference
2020	3rd Rank , fastMRI Challenge 2020, Facebook AI Research & NYU Langone Health	Virtual Conference
2020	ISMRM Summa Cum Laude , The ISMRM 28th Annual Meeting	Virtual Conference
2020	ISMRM The Poster Award of 2nd Place (Silver) , 2020 ISMRM Workshop on Data Sampling & Image Reconstruction	Sedona, US
2019	4th Rank , fastMRI Challenge 2019, Facebook AI Research & NYU Langone Health	Vancouver, Canada
2017	ISMRM Summa Cum Laude , The ISMRM 25th Annual Meeting	Hawaii, US

DOMESTIC

2026	MGB Radiology Innovation Award , Massachusetts General Brigham (MGB)	Boston, US
2021	Excellence Award , Medical Artificial Intelligence Datathon 2021, Ministry of Science and ICT and National Information Society Agency	Seoul, S.Korea
2021	Excellence Award , Hackathon of Development of AI-based Image Diagnosis using Medical Big Data 2021, Korea Testing Laboratory (KTL)	Seoul, S.Korea
2021	Best Paper Award , Graduate Student Paper Award, Yonsei University	Seoul, S.Korea
2019	Participation Prize , Samsung Humantech Paper Award (first author)	Seoul, S.Korea
2019	1st Rank and Grand Prize , HeLP Challenge 2018, Brain Tumor Segmentation Contest	Seoul, S.Korea
2018	Participation Prize , Samsung Humantech Paper Award (co-author)	Seoul, S.Korea
2017	Grand Prize , Yonsei Junior Convergence Science	Seoul, S.Korea

Grants

Advanced Neurovascular Imaging Using 7T MRI to Assess Long-Term Effects of Proton Radiotherapy in Pediatric Brain Tumor Survivors	National Institutes of Health (NIH), National Cancer Institute (NCI)
Role: co-PI, Total Direct Cost: \$2,931,339, Total Funding: \$4,181,335	Pending (5th percentile)
• NIH R01CA309465 (PIs: <u>Y. Jun</u> , C. Jaimes)	
Rapid, Motion-Robust, and Low-Gadolinium MRI for Pediatric Brain Tumors	National Institutes of Health (NIH), National Institute of Biomedical Imaging and Bioengineering (NIBIB)
Role: co-PI, Total Direct Cost: \$400,000, Total Funding: \$668,000	Sep. 2024 - Aug. 2027
• NIH R21EB036105 (PIs: <u>Y. Jun</u> , C. Jaimes)	

Scholarship

2023	ISMRM Trainee Stipend , ISMRM Workshop on Data Sampling and Image Reconstruction	US
2021	Dissertation Fellowship , Graduate Students Idea Incubation Fund, Yonsei University	S.Korea
2021	Academy Research Fellowship , Graduate Students Idea Incubation Fund, Yonsei University	S.Korea
2021	Best Paper Award Scholarship , Graduate Student Paper Award, Yonsei University	S.Korea
2020	ISMRM Trainee Stipend , ISMRM Workshop on Data Sampling and Image Reconstruction	US
2017-2019	ISMRM Educational Stipend , ISMRM	US
2019	Brain Korea 21 Plus Outstanding Student Fellow Scholarship , Korea Research Foundation	S.Korea
2018	Teaching Assistant Scholarship , Yonsei Univeristy	S.Korea
2017-2020	Brain Korea 21 Plus Scholarship , Korea Research Foundation	S.Korea
2016	Research Assistant Scholarship , Yonsei Univeristy	S.Korea
2012-2015	National Scholarship for Science & Engineering , Korea Student Aid Foundation	S.Korea

Invited Talks

From Quantitative to Diffusion MRI: Advanced Neuroimaging Driven by AI KU Graduate Program for Convergence & Translational Biomedicine • Korea University, Brain Convergence Research Center	Seoul, S.Korea Aug. 2025
AI Improvement in Image Quality and Analysis in Research and Clinical Practice ISMRM Pediatric MR Study Group Member-initiated Symposium • International Society for Magnetic Resonance in Medicine (ISMRM)	Hawaii, US May. 2025
Advanced neuroimaging using MRI: from quantitative MRI to diffusion MRI Brain Korea (BK) 21 Y-BASE R&E Institute • Yonsei University, School of Electrical and Electronic Engineering	Seoul, S.Korea May. 2024
Self-Supervised Learning for Rapid Quantitative MRI Athinoula A. Martinos Center for Biomedical Imaging • Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital	Boston, US May. 2023
Deep Model-based MR Parameter Mapping Network (DOPAMINE) for Fast MR Reconstruction 34th KSIIM Conference, 2020 • Korean Society of Imaging Informatics in Medicine	Seoul, S.Korea Oct. 2020
Medical Imaging Research using Artificial Intelligence HUFS AIM LAB, 2020 • The Catholic University of Korea, Eunpyeong St. Mary's Hospital	Seoul, S.Korea Jan. 2020

Presented Talks

Efficient mesoscale multiparametric quantitative MRI using 3D-QALAS at 7T with self-supervised learning ISMRM Annual Meeting, 2025 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2025	Hawaii, US May. 2025
Phase Reversed Interleaved Multi-Echo (PRIME) with phase, field map and motion navigators for highly accelerated distortion-free diffusion MRI ISMRM Annual Meeting, 2025 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2025	Hawaii, US May. 2025
PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI ISMRM Annual Meeting, 2024 • International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2024	Singapore May. 2024

Rapid Pediatric Imaging with Zero-Shot Deep Subspace Reconstruction for Multiparametric Quantitative MRI ISMRM Annual Meeting, 2024 International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2024	Singapore May. 2024
Zero-DeepSub: Zero-Shot Deep Subspace Reconstruction for Multiparametric Quantitative MRI Using QALAS ISMRM Annual Meeting, 2023 International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2023	Toronto, Canada June. 2023
Deep Subspace Reconstruction with Zero-Shot Learning for Multiparametric Quantitative MRI ISMRM Workshop on Data Sampling and Image Reconstruction, 2023 International Society for Magnetic Resonance in Medicine (ISMRM) on Data Sampling and Image Reconstruction, 2023	Sedona, US Jan. 2023
Joint Reconstruction of MR Image and Coil Sensitivity Maps using Deep Model-based Network ISMRM Annual Meeting, 2021 International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2021	Virtual Conference May. 2021
Deep Learning-based Automatic Detection and Segmentation of Brain Metastases Using Multi-Task Learning with 3D Black-Blood and GRE Imaging ISMRM Annual Meeting, 2021 International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2021	Virtual Conference May. 2021
Deep Model-based MR Parameter Mapping Network (DOPAMINE) for Fast MR Reconstruction ISMRM Annual Meeting, 2020 International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2020	Virtual Conference Aug. 2020
Deep Convolutional Neural Network for Acceleration of Magnetic Resonance Angiography (MRA) ISMRM Annual Meeting, 2017 International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting, 2017	Hawaii, US Apr. 2017

Publications - Preprints

Omni-QALAS: Optimized Multiparametric Imaging for Simultaneous T1, T2 and Myelin Water Mapping S Li, UD Gallastegi, S Fujita, Y Chen, P Xu, Y Choi, B Gagoski, H Ye, H Liu*, B Bilgic*, <u>Y Jun*</u> *Co-corresponding Authors, <i>arXiv preprint arXiv:2510.13118</i>	2025
Explainable AI for Accelerated Microstructure Imaging: A SHAP-Guided Protocol on the Connectome 2.0 scanner Q Uhl, T Pavan, J Gerold, KS Chan, <u>Y Jun</u>, S Fujita, A Bhatt, Y Ma, Q Wang, HH Lee, SY Huang, B Bilgic, I Jelescu <i>arXiv preprint arXiv:2509.09513</i>	2025
MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI Y Chen, <u>Y Jun</u>, A Heydari, X Yong, J Kim, J Lee, H Liu, H Ye, B Gagoski, S Fujita*, B Bilgic* <i>arXiv preprint arXiv:2508.10184</i>	2025
Multishot Dual Polarity GRAPPA: Robust Nyquist Ghost Correction for multishot EPI Y Jiang, <u>Y Jun</u>, Q Liu, W Zhong, Y Rath, H Guo, B Bilgic <i>arXiv preprint arXiv:2507.18273</i>	2025
A Tutorial on MRI Reconstruction: From Modern Methods to Clinical Implications T Çukur, SUH Dar, VA Nezhad, <u>Y Jun</u>, TH Kim, S Fujita, B Bilgic <i>arXiv preprint arXiv:2507.16715</i>	2025

- Evaluation of Synthetic Pediatric Brain MRI Using 3D-QALAS and Zero-DeepSub Reconstruction 2025
 SF Ferracioli[†], **Y Jun^{*}**, SAV Vasquez, VP Trujillo, H Griffin, S Fujita, E Milshteyn, B Bilgic, C Jaimes
 • [†] **Co-first Authors**, *research square preprint rs-7025131/v1*
- PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI 2024
Y Jun^{*}, Q Liu, T Gong, J Cho, S Fujita, X Yong, C Liao, ME Schmidt, S Nasr, C Jaimes, MS Gee, SY Huang, L Ning, A Yendiki, Y Rath, B Bilgic
 • *arXiv preprint arXiv:2409.07375*
- NLCG-Net: A Model-Based Zero-Shot Learning Framework for Undersampled Quantitative MRI Reconstruction 2024
 X Jian, **Y Jun**, J Cho, M Gao, X Yong, B Bilgic
 • *arXiv preprint arXiv:2401.12004*
- Improved Multi-Shot Diffusion-Weighted MRI with Zero-Shot Self-Supervised Learning Reconstruction 2023
 J Cho, **Y Jun**, X Wang, C Kobayashi, B Bilgic
 • *arXiv preprint arXiv:2308.05103*
- Zero-DeepSub: Zero-Shot Deep Subspace Reconstruction for Rapid Multiparametric Quantitative MRI Using 3D-QALAS 2023
Y Jun^{*}, Y Arefeen, J Cho, S Fujita, X Wang, PE Grant, B Gagoski, C Jaimes, MS Gee[#], B Bilgic[#]
 • ^{*} **Corresponding Author**, *arXiv preprint arXiv:2307.01410*
- SDC-UDA: Volumetric Unsupervised Domain Adaptation Framework for Slice-Direction Continuous Cross-Modality Medical Image Segmentation 2023
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *arXiv preprint arXiv:2305.11012*
- SSL-QALAS: Self-Supervised Learning for Rapid Multiparameter Estimation in Quantitative MRI Using 3D-QALAS 2023
Y Jun^{*}, J Cho, X Wang, M Gee, PE Grant, B Bilgic[#], B Gagoski[#]
 • ^{*} **Corresponding Author**, *arXiv preprint arXiv:2302.14240*
- COSMOS: Cross-Modality Unsupervised Domain Adaptation for 3D Medical Image Segmentation based on Target-aware Domain Translation and Iterative Self-Training 2022
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *arXiv preprint arXiv:2203.16557*
- Self-Training Based Unsupervised Cross-Modality Domain Adaptation for Vestibular Schwannoma and Cochlea Segmentation 2021
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *arXiv preprint arXiv:2109.10674*
- Results of the 2020 fastMRI Challenge for Machine Learning MR Image Reconstruction 2020
 MJ Muckley[†], B Riemenschneider[†], A Radmanesh, S Kim, G Jeong, J Ko, **Y Jun**, H Shin, D Hwang, M Mostapha, S Arberet, D Nickel, Z Ramzi, P Ciuciu, JL Starck, J Teuwen, D Karkalousos, C Zhang, A Sriram, Z Huang, N Yakubova, YW Lui, Florian Knoll
 • *arXiv preprint arXiv:2012.06318*

Publications - Peer-Review Journal

Omni-QALAS: Optimized Multiparametric Imaging for Simultaneous T1, T2 and Myelin Water Mapping

2026

S Li, UD Gallastegi, S Fujita, Y Chen, P Xu, KS Chan, Y Choi, J Kim, GF Piredda, T Hilbert, J Lee, H Ye, B Gagoski, H Liu*, B Bilgic*, **Y Jun***

- ***Co-corresponding Authors**, (under revision)

Vendor-Agnostic Joint Relaxometry and Myelin Water Fraction Mapping with B1 and Motion Correction

2026

S Fujita, **Y Jun**, AD Klauser, GF Piredda, T Hilbert, C Ariyurek, E Milshteyn, S Li, Y Chen, X Yong, KS Chan, Q Liu, S Yee, Y Rath, M Zaitsev, JF Nielsen, O Afacan, C Jaimes, PE Grant, B Gagoski#, B Bilgic#

- *Imaging Neuroscience*, (under revision)

Explainable AI for Accelerated Microstructure Imaging: A SHAP-Guided Protocol on the Connectome 2.0 scanner

2026

Q Uhl, T Pavan, J Gerold, KS Chan, **Y Jun**, S Fujita, A Bhatt, Y Ma, Q Wang, HH Lee, SY Huang, B Bilgic#, I Jelescu#

- *Imaging Neuroscience*, (under revision)

PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI

2026

Y Jun*, Q Liu, T Gong, J Cho, S Fujita, X Yong, C Liao, ME Schmidt, S Nasr, C Jaimes, MS Gee, SY Huang, L Ning, A Yendiki, Y Rath, B Bilgic

- ***Corresponding Author**, *Medical Image Analysis*, 111:104058

3D-QALAS synthetic MRI with Zero-DeepSub in children: initial experience including post-contrast imaging feasibility

2026

SF Ferracioli*, **Y Jun***, SAV Vasquez, VP Trujillo, H Griffin, S Fujita, E Milshteyn, B Bilgic, C Jaimes

- **+Co-first Authors, *Co-corresponding Authors**, *Pediatric Radiology*, 56(4):867-877

Multishot Dual Polarity GRAPPA: Robust Nyquist Ghost Correction for multishot EPI

2026

Y Jiang, **Y Jun**, Q Liu, W Zhong, Y Rath, H Guo, B Bilgic

- *Magnetic Resonance in Medicine*, 95(5):2726-2736

MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI

2026

Y Chen, **Y Jun**, A Heydari, X Yong, J Kim, J Lee, H Liu, H Ye, B Gagoski, S Fujita#, B Bilgic#

- *Magnetic Resonance in Medicine*, 95(3):1528-1544

A Tutorial on MRI Reconstruction: From Modern Methods to Clinical Implications

2025

T Çukur, SUH Dar, VA Nezhad, **Y Jun**, TH Kim, S Fujita, B Bilgic

- *IEEE Transactions on Biomedical Engineering*

Vendor-agnostic 3D multiparametric relaxometry improves cross-platform reproducibility

2025

S Fujita, B Gagoski, JF Nielsen, M Zaitsev, **Y Jun**, J Cho, X Yong, Q Uhl, P Xu, E Milshteyn, S Imam, Q Liu, Q Chen, O Afacan, JE Kirsch, Y Rath, B Bilgic

- *Magnetic Resonance in Medicine*, 94(3):937-948

Beyond the Conventional Structural MRI: Clinical Application of Deep Learning Image Reconstruction and Synthetic MRI of the Brain

2025

Y Choi, JS Ko, JE Park, G Jeong, M Seo, **Y Jun**, S Fujita, B Bilgic

- *Investigative Radiology*, 60(1):27-42

Zero-DeepSub: Zero-Shot Deep Subspace Reconstruction for Rapid Multiparametric Quantitative MRI Using 3D-QALAS <u>Y Jun</u> [*] , Y Arefeen, J Cho, S Fujita, X Wang, PE Grant, B Gagoski, C Jaimes, MS Gee [#] , B Bilgic [#] • [*] Corresponding Author , <i>Magnetic Resonance in Medicine</i> , 91(6):2459-2482	2024
SSL-QALAS: Self-Supervised Learning for Rapid Multiparameter Estimation in Quantitative MRI Using 3D-QALAS <u>Y Jun</u> [*] , J Cho, X Wang, M Gee, PE Grant, B Bilgic [#] , B Gagoski [#] • [*] Corresponding Author , <i>Magnetic Resonance in Medicine</i> , 90(5):2019-2032	2023
Deep learning referral suggestion and tumour discrimination using explainable artificial intelligence applied to multiparametric MRI H Shin, JE Park, <u>Y Jun</u> , T Eo, J Lee, JE Kim, DH Lee, HH Moon, SI Park, S Kim, D Hwang, HS Kim • <i>European Radiology</i> , 33:5859–5870	2023
Intelligent Noninvasive Meningioma Grading with a Fully Automatic Segmentation using Interpretable Multiparametric Deep Learning <u>Y Jun</u> ⁺ , YW Park ⁺ , H Shin ⁺ , Y Shin, JR Lee, K Han, SS Ahn, SM Lim, D Hwang, SK Lee • ⁺ Co-first Authors , <i>European Radiology</i> , 33(9):6124-6133	2023
Ultrathin crystalline-silicon-based strain gauges with deep learning algorithms for silent speech interfaces T Kim ⁺ , Y Shin ⁺ , K Kang ⁺ , K Kim ⁺ , G Kim ⁺ , Y Byeon ⁺ , H Kim, Y Gao, JR Lee, G Son, T Kim, <u>Y Jun</u> , J Kim, J Lee, S Um, Y Kwon, BG Son, M Cho, M Sang, J Shin, K Kim, J Suh, H Choi, S Hong, H Cheng, HG Kang [*] , D Hwang [*] , KJ Yu [*] • <i>Nature Communications</i> , 13:5815	2022
Results of the 2020 fastMRI Challenge for Machine Learning MR Image Reconstruction MJ Muckley ⁺ , B Riemschneider ⁺ , A Radmanesh, S Kim, G Jeong, J Ko, <u>Y Jun</u> , H Shin, D Hwang, M Mostapha, S Arberet, D Nickel, Z Ramzi, P Ciuciu, JL Starck, J Teuwen, D Karkalousos, C Zhang, A Sriram, Z Huang, N Yakubova, YW Lui, Florian Knoll • <i>IEEE Transactions on Medical Imaging</i> , 40(9):2306-2317	2021
Deep model-based magnetic resonance parameter mapping network (DOPAMINE) for fast T1 mapping using variable flip angle method <u>Y Jun</u> , H Shin, T Eo, T Kim, D Hwang • <i>Medical Image Analysis</i> , 70:102017	2021
Robust performance of deep learning for automatic detection and segmentation of brain metastases using three-dimensional black-blood and three-dimensional gradient echo imaging YW Park ⁺ , <u>Y Jun</u> ⁺ , Y Lee, K Han, C An, SS Ahn, D Hwang, SK Lee • ⁺ Co-first Authors , <i>European Radiology</i> , 31:6686-6695	2021
The Latest Trends in Attention Mechanisms and Their Application in Medical Imaging H Shin, J Lee, T Eo, <u>Y Jun</u> , S Kim, D Hwang • <i>Journal of the Korean Society of Radiology</i> , 81(6):1305-1333	2020
Accelerating Cartesian MRI by domain-transform manifold learning in phase-encoding direction T Eo ⁺ , H Shin ⁺ , <u>Y Jun</u> , T Kim, D Hwang • <i>Medical Image Analysis</i> , 63:101689	2020

- Parallel imaging in time-of-flight magnetic resonance angiography using deep multistream convolutional neural networks 2019
Y Jun, T Eo, H Shin, T Kim, HJ Lee, D Hwang
 • *Magnetic Resonance in Medicine*, 81(6):3840-3853
- Megahertz-wave-transmitting conducting polymer electrode for device-to-device integration 2019
 T Kim, G Kim, H Kim, HJ Yoon, T Kim, **Y Jun**, TH Shin, S Kang, J Cheon, D Hwang, BW Min, W Shim
 • *Nature Communications*, 10:653
- Deep-learned 3D black-blood imaging using automatic labelling technique and 3D convolutional neural networks for detecting metastatic brain tumors 2018
Y Jun, T Eo, T Kim, H Shin, D Hwang, SH Bae, YW Park, HJ Lee, BW Choi, SS Ahn
 • *Scientific Reports*, 8:9450
- KIKI-net: cross-domain convolutional neural networks for reconstructing undersampled magnetic resonance images 2018
 T Eo, **Y Jun**, T Kim, J Jang, HJ Lee, D Hwang
 • *Magnetic Resonance in Medicine*, 80(5):2188-2201
- High-SNR multiple T2 (*)-contrast magnetic resonance imaging using a robust denoising method based on tissue characteristics 2017
 T Eo, T Kim, **Y Jun**, H Lee, SS Ahn, DH Kim, D Hwang
 • *Journal of Magnetic Resonance Imaging*, 45(6):1835-1845

Publications - Conference Papers

- Improved Multi-Shot Diffusion-Weighted MRI with Zero-Shot Self-Supervised Learning Reconstruction 2023
 J Cho, **Y Jun**, X Wang, C Kobayashi, B Bilgic
 • *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp.457-466
- SDC-UDA: Volumetric Unsupervised Domain Adaptation Framework for Slice-Direction Continuous Cross-Modality Medical Image Segmentation 2023
 H Shin, H Kim, S Kim, **Y Jun**, T Eo, D Hwang
 • *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp.7412-7421
- Evaluation of the Robustness of Learned MR Image Reconstruction to Systematic Deviations Between Training and Test Data for the Models from the fastMRI Challenge 2021
 PM Johnson, G Jeong, K Hammernik, J Schlemper, C Qin, J Duan, D Rueckert, J Lee, N Pezzoti, ED Weerdt, S Yousefi, MS Elmahdy, JHFV Gemert, C Schuelke, M Doneva, T Nielsen, S Kastrayulin, BPF Lelieveldt, MJPV Osch, M Staring, EZ Chen, P Wang, X Chen, T Chen, VM Patel, S Sun, H Shin, **Y Jun**, T Eo, S Kim, T Kim, D Hwang, P Putzky, D Karkalousos, J Teuwen, N Miriakov, B Bakker, M Caan, M Welling, MJ Muckley, F Knoll
 • *International Workshop on Machine Learning for Medical Image Reconstruction (MLMIR)*, pp. 25-34
- Joint Deep Model-based MR Image and Coil Sensitivity Reconstruction Network (Joint-ICNet) for Fast MRI 2021
Y Jun, H Shin, T Eo, D Hwang
 • *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 5266-5275

Translation of 1D Inverse Fourier Transform of K-space to an Image Based on Deep Learning for Accelerating Magnetic Resonance Imaging 2018
 T Eo, H Shin, T Kim, **Y Jun**, D Hwang
 • *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 241-249

Publications - Conference Abstracts

Efficient mesoscale multiparametric quantitative MRI using 3D-QALAS at 7T with self-supervised learning 2025

Y Jun, S Fujita, Yu Chen, A Mareyam, C Jaimes, MS Gee, B Gagoski, B Bilgic
 • **[*Oral Presentation]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.0815

Phase Reversed Interleaved Multi-Echo (PRIME) with phase, field map and motion navigators for highly accelerated distortion-free diffusion MRI 2025

Y Jun, Q Liu, T Gong, J Cho, S Fujita, X Yong, SY Huang, L Ning, A Yendiki, Y Rath, B Bilgic
 • **[*Oral Presentation]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.0514

MIMOSA: Multi-parametric Imaging using Multiple-echoes with Optimized Simultaneous Acquisition for highly-efficient quantitative MRI 2025

Y Chen, **Y Jun**, A Heydari, X Yong, H Liu, H Ye, B Gagoski, B Bilgic, S Fujita
 • **[*Oral Presentation]** **[*Magna Cum Laude]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.0812

Robust Nyquist ghost correction for high-resolution EPI using multishot dual-polarity GRAPPA reconstruction 2025

Y Jiang, **Y Jun**, Q Liu, W Zhong, Y Rath, H Guo, B Bilgic
 • **[*Oral Presentation]** **[*Magna Cum Laude]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.1365

Vendor-Agnostic Joint Relaxometry and Myelin Water Fraction Mapping with B1 Correction 2025

S Fujita, **Y Jun**, AD Klauser, GF Piredda, T Hilbert, C Ariyurek, E Milshteyn, Q Liu, IA Shaik, Y Rath, M Zaitsev, JF Nielsen, C Jaimes, PE Grant, O Afacan, B Gagoski, B Bilgic
 • **[*Oral Presentation]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.1104

Mesoscale Myelin Water Fraction Mapping at 3T with Self-navigated Motion Correction 2025

S Fujita, **Y Jun**, AD Klauser, GF Piredda, T Hilbert, C Ariyurek, O Afacan, B Gagoski, B Bilgic
 • **[*Oral Presentation]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.1109

Reducing the NEXI acquisition time for the quantification of human gray matter microstructure on the CONNECTOM 2.0 scanner 2025

Q Uhl, T Pavan, J Gerold, KS Chan, **Y Jun**, A Bhatt, Y Ma, HH Lee, SY Huang, B Bilgic, I Jelescu
 • **[*Oral Presentation]** **[*Summa Cum Laude]** *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.0129

vNav-QALAS: Motion robust 3D multi-parametric brain mapping with volumetric navigators 2025

P Xu, S Fujita, **Y Jun**, B Gagoski, O Afacan, H Liu, B Bilgic
 • *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.4433

Characterization of human brain IVIM signal using two-dimensional T2-diffusivity spectrum analysis based on multi-echo diffusion MRI 2025

Z Hu, D Varadarajan, **Y Jun**, GA Hartung, A Arsenovic, LD Lewis, SY Huang, KM Kwong, B Bilgic, B Rosen, JR Polimeni
 • *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.2029

Zero-Shot Self-Supervised Distortion-Free Diffusion MRI Reconstruction 2025

MY Avci, J Cho, **Y Jun**, B Bilgic
 • *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.4807

Motion-Robust T1/T2 Mapping of the Abdomen using Pilot-Tone Navigation 2025

C Ariyurek, B Bilgic, S Fujita, **Y Jun**, S Kurugol, B Gagoski, O Afacan
 • *International Society for Magnetic Resonance in Medicine (ISMRM)*, pp.5138

PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI	2024
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Patents

Phase Reversed Interleaved Multi-Echo (PRIME) acquisition for highly accelerated distortion-free diffusion MRI	2025
B Bilgic, J Cho, S Fujita, Y Jun , Q Liu, Y Rath, X Yong	
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Method And Device For Correcting Medical Image Using Phantom	2022
D Hwang, Y Jun , T Kim, Y Lee	
• <i>Registered, 10-2481027, (S. Korea)</i>	
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Activities

Ad Hoc Reviewer

- IEEE Transactions on Medical Imaging (IEEE TMI) (*Distinguished Reviewer)
- Magnetic Resonance in Medicine
- Medical Physics
- Artificial Intelligence in Medicine
- Scientific Reports
- Imaging Neuroscience
- Neuroradiology
- Engineering Applications of Artificial Intelligence
- Frontiers in Pediatrics
- Quantitative Imaging in Medicine and Surgery
- BMC Medical Imaging
- IEEE Journal of Biomedical and Health Informatics (IEEE JBHI)
- IEEE Access
- IEEE Sensors Letters
- International Society for Magnetic Resonance in Medicine (ISMRM 2022-2026)
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2020-2025)
- European Conference on Computer Vision (ECCV 2024)
- International Conference on Computer Vision (ICCV 2025)
- Conference on Computer Vision and Pattern Recognition (CVPR 2025-2026)

Conference Moderator

- Moderator of International Society for Magnetic Resonance in Medicine (ISMRM 2024-2025)
- Poster Facilitator of International Society for Magnetic Resonance in Medicine (ISMRM 2021)

Professional Societies

- Full Member of International Society for Magnetic Resonance in Medicine (ISMRM) (2025-)
- Trainee Member of International Society for Magnetic Resonance in Medicine (ISMRM) (2017-2024)
- ISMRM Study Groups (Quantitative MRI, Ultra-high Field MR, Diffusion, Pediatric MR)

Skills

Programming	Python, Matlab, Pytorch, Tensorflow/Keras, C/C++
MRI Sequence Design	Pulseseq
Languages	Korean, English

References

Available upon request